FCFS Simulation Brief: Main Findings and Discussion Questions

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Executive Summary

Under this FCFS stress-test baseline, utilization should not be used as the access summary; report served rate, offered wait, and no-offer share together. Across the 30-seed baseline diagnostic ($\lambda = 100$ arrivals/day, $S = 32$ slots/day), average utilization is 0.842 while only 0.269 of arriving patients complete a visit. These metrics have different denominators and respond to different interventions. Utilization alone understates the access shortfall at high demand.

- **Utilization and access decouple by construction.** With $S = 32$ slots and $\lambda = 100$ arrivals, the served rate is mechanically bounded near $32/100 = 0.32$ before behavioral losses. Slot fill can be high while most patients are turned away.
- **Baseline losses are post-offer, not no-offer.** 26.9% of arrivals are served; 35.5% balk, 32.5% cancel, 5.1% no-show, and 0.0% receive no offer. The horizon does not saturate at baseline. Slots are offered, then rejected, canceled, or missed.
- **Suppressing balking does not recover capacity.** Removing balking shifts utilization by -0.001 and served rate by -0.001. Losses relabel from balking to no-offer and cancellation. The completed-visit count is unchanged.

Caveat: This is a stress-test model, not a calibrated clinic estimate. The $3.1 : 1$ *demand-to-capacity ratio is intentionally severe.*

Discussion questions for next meeting:

1. Should the baseline be recalibrated toward a plausible clinic setting, or does the stress-test regime remain useful for probing mechanism?
2. Should no-offer be reported as a distinct access failure in every summary table?
3. Should no-show probability depend on original offered delay, as currently implemented, or on remaining wait closer to the service date?

Key Evidence

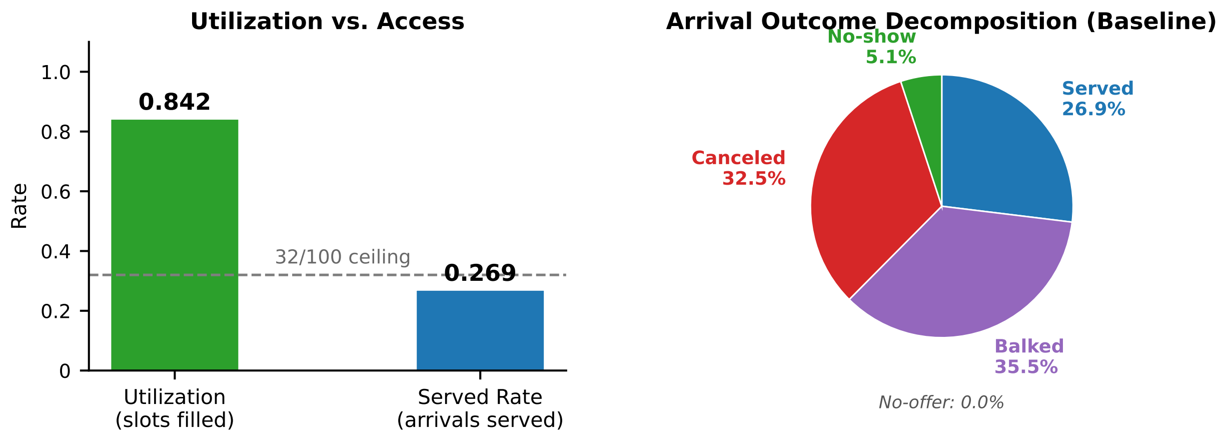

Figure 1: Left: utilization vs. served rate at baseline. The dashed line marks the mechanical ceiling $32/100 = 0.32$. Right: arrival outcome decomposition over 30 seeds ($\lambda = 100$, $S = 32$).

Metric	Value	Notes
Average utilization	0.842	Completed visits per available slot
Overall served rate	0.269	Completed visits per arrival
Mean offered wait	9.29 d	Delay among patients offered a slot
Mean accepted wait	8.33 d	Delay among patients who accepted
Overall balking rate	0.355	Offered patients who rejected the delay

Table 1: Baseline FCFS metrics (30 seeds).

Metric	Baseline	No-balking	Change
Average utilization	0.842	0.840	−0.001
Overall served rate	0.269	0.268	−0.001
Booked rate	0.645	0.703	+0.058
Balked share	0.355	0.000	−0.355
Canceled share	0.325	0.384	+0.059
No-offer share	0.000	0.297	+0.297
Mean accepted wait	8.33 d	9.45 d	+1.12 d

Table 2: No-balking diagnostic, 30 matched seeds. Utilization and served rate are essentially unchanged; losses relabel.

Main Takeaways

Utilization Is Not Access

Utilization measures completed visits per available slot; served rate measures completed visits per arrival. At $\lambda = 100$ and $S = 32$, the served-rate ceiling is $32/100 = 0.32$ before any behavioral losses. The two metrics can move independently: a policy that fills more slots does not necessarily serve more patients, and vice versa. Using utilization as a proxy for access will be misleading whenever the demand-to-capacity ratio is far from one.

Baseline Losses Are Balking and Cancellation

Every arrival maps to exactly one outcome. At baseline, no-offer is zero: the system finds a future slot for every arriving patient. Access failures occur after offers are made, not because the booking horizon is already unavailable at baseline. Balking (35.5%) reflects high offered delay; cancellation (32.5%) reflects long exposure before service combined with a daily cancellation probability. The capacity pressure is severe, but the baseline loss channel is post-offer dropout, not slot unavailability.

Suppressing Balking Does Not Recover Capacity

Table 2 is the cleanest diagnostic. Removing balking raises the booked rate by 0.058 but leaves utilization and served rate essentially unchanged. The additional bookings extend far into the horizon, accumulate cancellation exposure, and generate no-offer arrivals once the horizon fills. Suppressing balking does not recover capacity. It relabels losses.

Report Offered Wait with No-Offer Share

Accepted wait is a selective metric: patients who balk at long delays are excluded. As congestion rises, accepted wait can decrease simply because high-delay patients opt out, even as access worsens. Offered wait is a better congestion measure because it is recorded before the balking decision. However, offered wait excludes no-offer patients. At baseline, no-offer is zero; but when balking is suppressed, no-offer rises to 0.297 and offered wait becomes selective in a different way. Report offered wait and no-offer share together.

Realistic Scenario Check

A comparison run under a plausible two-class parameterization ($\lambda = 24$, $S = 20$, demand/supply $1.2 : 1$, asymmetric behavioral parameters; 30 seeds) shows what changes when capacity pressure is reduced. Aggregate outcomes improve substantially: utilization 0.934, served rate 0.789, balked share 3.3%, canceled share 12.3%, no-offer share 0.0%. The critical new result is a class-level served-rate gap of 0.220 (Class 1: 0.881, Class 2: 0.661) that is invisible in the aggregate figure. Under pooled FCFS, asymmetric behavioral parameters create this gap through shared capacity, not explicit class priority. The reporting recommendation is unchanged: aggregate metrics alone are insufficient; class-level served rates must also be reported.

Driver Hierarchy

Sensitivity slices and regression screen give a consistent hierarchy; all coefficients are standardized OLS with HC3 errors.

- **Served rate and offered wait:** driven by arrival rate ($\hat{\beta} = -0.774$ and $+0.576$; $p < 0.001$). Demand pressure dominates when capacity is fixed.
- **Utilization:** driven by no-show behavior ($\hat{\beta} = +0.512$ threshold, -0.423 step; $p < 0.001$). No-show slots are not rebooked.
- **Balking effect on utilization:** near zero ($\hat{\beta} = +0.016$, $p = 0.72$). Balking reallocates loss channels, not aggregate slot completion.
- **Class access gap:** requires asymmetric parameters. Symmetric baseline yields a gap of 0.001; the realistic scenario yields 0.220 (see above).

Open Questions

1. **Recalibrate or keep stress test?** The realistic scenario ($\lambda/S = 1.2$) provides a first check: aggregate outcomes improve and the loss-channel mix shifts. The stress-test baseline remains useful for mechanism diagnosis. The open question is which parameter regime should serve as the main reference case, and what clinical inputs justify that choice.
2. **Report no-offer as distinct access failure?** Zero at baseline, but appears immediately under higher λ or without balking. Including it in every summary table prevents silent omissions when parameter regimes change.
3. **No-show rule: original delay or residual wait?** The current rule uses original booking delay, not residual wait. Switching to residual wait would make offered delay a more direct driver of utilization and would likely alter the regression hierarchy.

Where to Dig

This brief stands alone. The table below points to the full note (`fcfs_metric_analysis_note.pdf`) for details.

Topic	Full-note reference
Baseline setup and metric definitions	Sections 2.1–2.3
Utilization vs. served-rate gap	Section 2.3; Section 5
Outcome decomposition detail	Section 3.1, Table 2
No-balking diagnostic	Section 3.2, Table 3
Sensitivity driver hierarchy	Sections 6.1–6.5
Regression coefficients (full table)	Appendix D
Class gap and equity	Sections 6.5, 8
Discussion and caveats	Sections 9–10
Realistic scenario comparison	`realistic_baseline_comparison.pdf`